



13.0.1 NIELIT 2018-27



The following vectors $(1, 9, 9, 8), (2, 0, 0, 8), (2, 0, 0, 3)$ are

- A. Linearly dependent
C. Constant
B. Linearly independent
D. None of these

nielit-2018 engineering-mathematics linear-algebra

Answer key

13.1

Determinant (8)

13.1.1 Determinant: NIELIT 2016 DEC Scientist B (CS) - Section B: 21



Let A, B, C, D be $n \times n$ matrices, each with non-zero determinant. If $ABCD = 1$, then B^{-1} is:

- A. $D^{-1}C^{-1}A^{-1}$
C. ADC
B. CDA
D. Does not necessarily exist.

nielit2016dec-scientistb-cs engineering-mathematics linear-algebra matrix determinant

Answer key

13.1.2 Determinant: NIELIT 2016 DEC Scientist B (IT) - Section B: 26



Two eigenvalues of a 3×3 real matrix P are $(2 + \sqrt{-1})$ and 3. The determinant of P is _____.

- A. 0
B. 1
C. 15
D. -1

nielit2016dec-scientistb-it engineering-mathematics linear-algebra determinant eigen-value

Answer key

13.1.3 Determinant: NIELIT 2016 MAR Scientist B - Section B: 12



If A and B are square matrices of size $n \times n$, then which of the following statements is not true?

- A. $\det(AB) = \det(A)\det(B)$
C. $\det(A+B) = \det(A) + \det(B)$
B. $\det(kA) = k^n \det(A)$
D. $\det(A^T) = 1/\det(A^{-1})$

nielit2016mar-scientistb engineering-mathematics linear-algebra determinant

Answer key

13.1.4 Determinant: NIELIT 2016 MAR Scientist B - Section B: 4



What is the determinant of the matrix

$$\begin{bmatrix} 5 & 3 & 2 \\ 1 & 2 & 6 \\ 3 & 5 & 10 \end{bmatrix}$$

- A. -76 B. -28 C. $+28$ D. $+72$

nielit2016mar-scientistb engineering-mathematics linear-algebra determinant

Answer key 

13.1.5 Determinant: NIELIT 2017 OCT Scientific Assistant A (CS) - Section C: 9

M is a square matrix of order ' n ' and its determinant value is 5. If all the elements of M are multiple by 2, its determinant value becomes 40. The value of ' n ' is



- A. 2 B. 3 C. 5 D. 4

nielit2017oct-assistanta-cs engineering-mathematics linear-algebra matrix determinant

Answer key 

13.1.6 Determinant: NIELIT 2017 OCT Scientific Assistant A (IT) - Section B: 7

M is a square matrix of order ' n ' and its determinant value is 5. If all the elements of M are multiplied by 2, its determinant value becomes 40. The value of ' n ' is



- A. 2 B. 3 C. 5 D. 4

nielit2017oct-assistanta-it linear-algebra matrix determinant

Answer key 

13.1.7 Determinant: NIELIT 2021 Dec Scientist B - Section B: 18

The determinant of matrix $\begin{bmatrix} 0 & p-q & p-r \\ q-p & 0 & q-r \\ r-p & r-q & 0 \end{bmatrix}$ is _____.



- A. 0 B. $(p-q)(q-r)(r-p)$
C. pqr D. $3pqr$

nielit-2021-it-dec-scientistb matrix determinant linear-algebra

Answer key 

13.1.8 Determinant: NIELIT 2021 Dec Scientist B - Section B: 58

The determinant of matrix $\begin{bmatrix} 0 & p-q & p-r \\ q-p & 0 & q-r \\ r-p & r-q & 0 \end{bmatrix}$ is _____.



A. 0

B. $(p - q)(q - r)(r - p)pqr$ D. $3pqr$

nielit2021dec-scientistb matrix determinant

Answer key 

13.2

Eigen Value (3)

13.2.1 Eigen Value: NIELIT 2016 MAR Scientist C - Section B: 8



The eigenvalues of the matrix $\begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix}$ are

A. 5 and -5

B. 5 and -1

C. 1 and -5

D. 2 and 3

nielit2016mar-scientistc engineering-mathematics linear-algebra eigen-value

Answer key 

13.2.2 Eigen Value: NIELIT 2018-21



The value of p such that the vector $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ is an eigen vector of the matrix

$$\begin{bmatrix} 4 & 1 & 2 \\ p & 2 & 1 \\ 14 & -4 & 10 \end{bmatrix} \text{ is}$$

A. 15

B. 16

C. 17

D. 18

nielit-2018 engineering-mathematics linear-algebra matrix eigen-value

Answer key 

13.2.3 Eigen Value: NIELIT 2022 April Scientist B | Section B | Question: 72



The Eigen values of a matrix

$A = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$ are $-3, -3$ and 5 , then the trace of the matrix $A^3 - 3A^2$ is:

A. 200

B. 71

C. -58 D. -200

nielit2022apr-scientistb linear-algebra matrix eigen-value

Answer key 

13.3

Eigen Vector (1)

13.3.1 Eigen Vector: NIELIT 2016 MAR Scientist C - Section B: 9



Consider three vectors $x = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, $y = \begin{bmatrix} 4 \\ 8 \end{bmatrix}$, $z = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$. Which of the following statements is true

- A. x and y are linearly independent
- B. x and y are linearly dependent
- C. x and z are linearly dependent
- D. y and z are linearly dependent

nielit2016mar-scientistc engineering-mathematics linear-algebra eigen-vector

Answer key

13.4

Matrix (5)

13.4.1 Matrix: NIELIT 2016 MAR Scientist C - Section B: 19



If product of matrix $A = \begin{bmatrix} \cos^2 \theta & \cos \theta \sin \theta \\ \cos \theta \sin \theta & \sin^2 \theta \end{bmatrix}$ and $B = \begin{bmatrix} \cos^2 \phi & \cos \phi \sin \phi \\ \cos \phi \sin \phi & \sin^2 \phi \end{bmatrix}$ is a null matrix, then θ and ϕ differ by an

- A. odd multiple of π
- B. even multiple of π
- C. odd multiple of $\frac{\pi}{2}$
- D. even multiple of $\frac{\pi}{2}$

nielit2016mar-scientistc linear-algebra matrix

13.4.2 Matrix: NIELIT 2016 MAR Scientist C - Section B: 3



The matrices $\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$ and $\begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$ commute under the multiplication

- A. if $a = b$ (or) $\theta = n\pi$, n is an integer
- B. always
- C. never
- D. if $a \cos \theta \neq b \sin \theta$

nielit2016mar-scientistc linear-algebra matrix

Answer key

13.4.3 Matrix: NIELIT 2017 DEC Scientist B - Section B: 60



Consider two matrices M_1 and M_2 with $M_1^* M_2 = 0$ and M_1 is non singular. Then which of the following is true?

- A. M_2 is non singular
- B. M_2 is null matrix
- C. M_2 is the identity matrix
- D. M_2 is transpose of M_1

nielit2017dec-scientistb engineering-mathematics linear-algebra matrix

Answer key

13.4.4 Matrix: NIELIT 2018-30



If C is a non-singular matrix and $B = C \begin{bmatrix} 0 & x & y \\ 0 & 0 & x \\ 0 & 0 & 0 \end{bmatrix} C^{-1}$ then:

- A. $B^2 = I$ B. $B^2 = \text{Null Matrix}$ C. $B^3 = I$ D. $B^3 = \text{Null Matrix}$

nielit-2018 engineering-mathematics linear-algebra matrix

Answer key

13.4.5 Matrix: NIELIT 2021 Dec Scientist A - Section B: 82



If for the matrix A , $A^3 = I$ then $A^{-1} =$ _____ .

- A. A^2 B. A^3 C. A D. None of these

nielit2021dec-scientista matrix linear-algebra

Answer key

13.5

System of Equations (3)

13.5.1 System of Equations: NIELIT 2016 MAR Scientist B - Section B: 6



The system of simultaneous equations

$$x + 2y + z = 6$$

$$2x + y + 2z = 6$$

$$x + y + z = 5$$

has

- A. unique solution. B. infinite number of solutions.
C. no solution. D. exactly two solutions.

nielit2016mar-scientistb engineering-mathematics linear-algebra system-of-equations

Answer key

13.5.2 System of Equations: NIELIT 2022 April Scientist B | Section B | Question: 80



The upper triangular matrix U in the LU- decomposition of the matrix given below :

$$\begin{bmatrix} 1 & -2 & 3 \\ 2 & -5 & 8 \\ 1 & 2 & -10 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix} \text{ is :}$$

A. $\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1 & -4 & 1 \end{bmatrix}$

B. $\begin{bmatrix} 1 & -2 & 3 \\ 0 & 1 & -6 \\ 0 & 0 & 1 \end{bmatrix}$

C. $\begin{bmatrix} 1 & -2 & 3 \\ 0 & -1 & 6 \\ 0 & 0 & 11 \end{bmatrix}$

D. $\begin{bmatrix} 1 & -2 & 3 \\ 0 & -1 & 6 \\ 0 & 0 & 10 \end{bmatrix}$

nielit2022apr-scientistb matrix linear-algebra system-of-equations

Answer key 

13.5.3 System of Equations: NIELIT Scientist B 2020 November: 27



What is the value of k for which the following system of equations has no solution:
 $2x-8y=3$ and $kx+4y=10$

- A. -2 B. 1 C. -1 D. 2

nielit-scb-2020 system-of-equations

Answer key 

Answer Keys

13.0.1	D	13.1.1	B	13.1.2	C	13.1.3	C	13.1.4	B
13.1.5	B	13.1.6	B	13.1.7	A	13.1.8	A	13.2.1	B
13.2.2	D	13.2.3	C	13.3.1	B	13.4.1	C	13.4.2	A
13.4.3	B	13.4.4	A	13.4.5	A	13.5.1	C	13.5.2	TBA
13.5.3	C								